KAIZEN MATHS

Lesson starter

Proof by Induction

Matrix Powers

Name _____

S1

Prove each matrix power formula by mathematical induction.

Let $A = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$. Prove by induction that $A^n = \begin{pmatrix} 1 & 3n \\ 0 & 1 \end{pmatrix}$ for $n \geq 1$.

S2

Prove each matrix power formula by mathematical induction.

Let $D = \begin{pmatrix} 3 & 0 \\ 0 & 6 \end{pmatrix}$. Prove by induction that $D^n = \begin{pmatrix} 3^n & 0 \\ 0 & 6^n \end{pmatrix}$ for $n \geq 1$.

S3

Prove each matrix power formula by mathematical induction.

Let $A = \begin{pmatrix} 3 & 3 \\ 0 & 3 \end{pmatrix}$. Prove by induction that $A^n = \begin{pmatrix} 3^n & 3n(3^n-1) \\ 0 & 3^n \end{pmatrix}$ for $n \geq 1$.

S4

Prove each matrix power formula by mathematical induction.

Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$. Prove by induction that $A^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$ for $n \geq 1$.

Show your working

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Let $D = \begin{pmatrix} 3 & 0 \\ 0 & 6 \end{pmatrix}$. Prove by induction that $D^n = \begin{pmatrix} 3^n & 0 \\ 0 & 6^n \end{pmatrix}$ for $n \geq 1$.

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Prove each matrix power formula by mathematical induction.

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KAIZEN MATHS

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Proof by Induction

Matrix Powers

Name _____

EA1

Prove each matrix power formula by mathematical induction.

Let $D = \begin{matrix} 2 & 0 \\ 0 & 5 \end{matrix}$. Prove by induction that $D^n = \begin{matrix} 2^n & 0 \\ 0 & 5^n \end{matrix}$ for $n \geq 1$.

EA2

Prove each matrix power formula by mathematical induction.

Let $A = \begin{matrix} 4 & 2 \\ 0 & 4 \end{matrix}$. Prove by induction that $A^n = \begin{matrix} 4^n & 2n(4^n-1) \\ 0 & 4^n \end{matrix}$ for $n \geq 1$.

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Proof by Induction

Matrix Powers

Name _____

EB1

Prove each matrix power formula by mathematical induction.

Let $A = \begin{matrix} 1 & 5 \\ 0 & 1 \end{matrix}$. Prove by induction that $A^n = \begin{matrix} 1 & 5n \\ 0 & 1 \end{matrix}$ for $n \geq 1$.

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Proof by Induction

Matrix Powers

Name _____

EA1

Prove each matrix power formula by mathematical induction.

Let $D = \begin{matrix} 2 & 0 \\ 0 & 5 \end{matrix}$. Prove by induction that $D^n = \begin{matrix} 2^n & 0 \\ 0 & 5^n \end{matrix}$ for $n \geq 1$.

EA2

Prove each matrix power formula by mathematical induction.

Let $A = \begin{matrix} 4 & 2 \\ 0 & 4 \end{matrix}$. Prove by induction that $A^n = \begin{matrix} 4^n & 2n(4^n-1) \\ 0 & 4^n \end{matrix}$ for $n \geq 1$.

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Proof by Induction

Matrix Powers

Name _____

EB1

Prove each matrix power formula by mathematical induction.

Let $A = \begin{matrix} 1 & 5 \\ 0 & 1 \end{matrix}$. Prove by induction that $A^n = \begin{matrix} 1 & 5n \\ 0 & 1 \end{matrix}$ for $n \geq 1$.

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Teacher answers & working

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S1

Prove each matrix power formula by mathematical induction.

Let A = 1 3 0 1 . Prove by induction that A^n = 1 3n 0 1 for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: $A^1 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$, which matches $\begin{bmatrix} 1 & 3(1) \\ 0 & 1 \end{bmatrix}$.

Assume true for n = k: $A^k = \begin{bmatrix} 1 & 3k \\ 0 & 1 \end{bmatrix}$.

Then $A^{k+1} = A^k A$.

$$A^{k+1} = \begin{bmatrix} 1 & 3k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 3 + 3k \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 3(k+1) \\ 0 & 1 \end{bmatrix}.$$

This is the required formula for n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers n ≥ 1.

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S2

Prove each matrix power formula by mathematical induction.

Let D = 3 0 0 6 . Prove by induction that $D^n = 3^n\ 0\ 0\ 6^n$ for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: $D^1 = \begin{bmatrix} 3 & 0 \\ 0 & 6 \end{bmatrix}$, which matches $\begin{bmatrix} 3^1 & 0 \\ 0 & 6^1 \end{bmatrix}$.

Assume true for n = k: $D^k = \begin{bmatrix} 3^k & 0 \\ 0 & 6^k \end{bmatrix}$.

Then $D^{k+1} = D^kD$.

$$D^{k+1} = \begin{bmatrix} 3^k & 0 \\ 0 & 6^k \end{bmatrix}\begin{bmatrix} 3 & 0 \\ 0 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 3^k \times 3 & 0 \\ 0 & 6^k \times 6 \end{bmatrix}$$

$$= \begin{bmatrix} 3^{k+1} & 0 \\ 0 & 6^{k+1} \end{bmatrix}.$$

This is the required formula for n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers n ≥ 1.

Teacher answers & working

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S3

Prove each matrix power formula by mathematical induction.

Let A = 3 3 0 3 . Prove by induction that A^n = 3^n 3n(3^n−1) 0 3^n for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: $A^1 = \begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix}$.

The formula gives $\begin{bmatrix} 3^1 & 3(1)(3^0) \\ 0 & 3^1 \end{bmatrix}$, which is the same matrix.

Assume true for n = k: $A^k = \begin{bmatrix} 3^k & 3k(3^{k-1}) \\ 0 & 3^k \end{bmatrix}$.

Then $A^{k+1} = A^kA$.

$$A^{k+1} = \begin{bmatrix} 3^k & 3k(3^{k-1}) \\ 0 & 3^k \end{bmatrix}\begin{bmatrix} 3 & 3 \\ 0 & 3 \end{bmatrix}$$

Top-right entry = $3^k(3) + 3k(3^{k-1})(3)$

$= 33^k + 3k3^k = 3(k+1)3^k$.

So $A^{k+1} = \begin{bmatrix} 3^{k+1} & 3(k+1)3^k \\ 0 & 3^{k+1} \end{bmatrix}$.

This is the required formula for n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers n ≥ 1.

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S4

Prove each matrix power formula by mathematical induction.

Let A = 1 2 0 1 . Prove by induction that A^n = 1 2n 0 1 for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: $A^1 = \begin{bmatrix}1 & 2\\0 & 1\end{bmatrix}$, which matches $\begin{bmatrix}1 & 2(1)\\0 & 1\end{bmatrix}$.

Assume true for n = k: $A^k = \begin{bmatrix}1 & 2k\\0 & 1\end{bmatrix}$.

Then $A^{k+1} = A^kA$.

$$A^{k+1} = \begin{bmatrix}1 & 2k\\0 & 1\end{bmatrix}\begin{bmatrix}1 & 2\\0 & 1\end{bmatrix}$$

$$= \begin{bmatrix}1 & 2+2k\\0 & 1\end{bmatrix}$$

$$= \begin{bmatrix}1 & 2(k+1)\\0 & 1\end{bmatrix}.$$

This is the required formula for n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers n ≥ 1.

Teacher answers & working

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EA1

Prove each matrix power formula by mathematical induction.

Let D = 2 0 0 5 . Prove by induction that $D^n = 2^n\ 0\ 0\ 5^n$ for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: $D^1 = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$, which matches $\begin{bmatrix} 2^1 & 0 \\ 0 & 5^1 \end{bmatrix}$.

Assume true for n = k: $D^k = \begin{bmatrix} 2^k & 0 \\ 0 & 5^k \end{bmatrix}$.

Then $D^{k+1} = D^kD$.

$$D^{k+1} = \begin{bmatrix} 2^k & 0 \\ 0 & 5^k \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 2^k \times 2 & 0 \\ 0 & 5^k \times 5 \end{bmatrix}$$

$$= \begin{bmatrix} 2^{k+1} & 0 \\ 0 & 5^{k+1} \end{bmatrix}.$$

This is the required formula for n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers n ≥ 1.

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EA2

Prove each matrix power formula by mathematical induction.

Let A = 4 2 0 4 . Prove by induction that $A^n = 4^n\ 2n(4^n-1)\ 0\ 4^n$ for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: $A^1 = \begin{bmatrix}4 & 2\\ 0 & 4\end{bmatrix}$.

The formula gives $\begin{bmatrix}4^1 & 2(1)(4^0)\\ 0 & 4^1\end{bmatrix}$, which is the same matrix.

Assume true for n = k: $A^k = \begin{bmatrix}4^k & 2k(4^{k-1})\\ 0 & 4^k\end{bmatrix}$.

Then $A^{k+1} = A^kA$.

$$A^{k+1} = \begin{bmatrix}4^k & 2k(4^{k-1})\\ 0 & 4^k\end{bmatrix}\begin{bmatrix}4 & 2\\ 0 & 4\end{bmatrix}$$

Top-right entry = $4^k(2) + 2k(4^{k-1})(4)$

$= 2 4^k + 2k4^k = 2(k+1)4^k$.

So $A^{k+1} = \begin{bmatrix}4^{k+1} & 2(k+1)4^k\\ 0 & 4^{k+1}\end{bmatrix}$.

This is the required formula for n = k + 1.

The base case has been shown to be true. The induction step has shown that if the statement is true for n = k, then it is true for n = k + 1. Therefore, by mathematical induction, the statement is true for all integers n ≥ 1.

Teacher answers & working

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EB1

Prove each matrix power formula by mathematical induction.

Let A = 1 5 0 1 . Prove by induction that A^n = 1 5n 0 1 for n ≥ 1.

Answer:

Proved by induction.

Working:

Base case n = 1: $A^1 = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$, which matches $\begin{bmatrix} 1 & 5(1) \\ 0 & 1 \end{bmatrix}$.

Assume true for n = k: $A^k = \begin{bmatrix} 1 & 5k \\ 0 & 1 \end{bmatrix}$.

Then $A^{k+1} = A^k A$.

$$A^{k+1} = \begin{bmatrix} 1 & 5k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 5 + 5k \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 5(k+1) \\ 0 & 1 \end{bmatrix}.$$

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